

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF MANAGEMENT STUDIES (FMS)
BBA PROGRAMME
SEMESTER – II, UNIVERSITY EXAMINATION, SEPTEMBER - 2018
BM110 BUSINESS STATISTICS (OLD)
BM110.1 BUSINESS STATISTICS (NEW)

Date: 20/09/2018, Thursday

Time: 10.00 AM to 10.30 AM

Total Marks: 15

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Important Instructions:

- All questions are compulsory.
- This is a closed book examination.
- There are two sections in the question paper. Section – I contains Q. No.: 1 which is of 15 marks and Section – II contains total five questions wherein Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
- Section – I should be answered in the given question paper and Section – II should be answered in a separate answer book.
- Maximum time allotted for Section – I is 30 minutes. Answer Book for Section – II will be given when the Question Paper for Section – I is returned. However, candidates who complete Section – I earlier than 30 minutes, can collect Answer Book for Section – II immediately after handing over Section – I question paper.
- Figures on the right indicate marks for each question.

SECTION – I

Q.1 Objective Type Questions:

15

A) Match the following: (i – vi)

06

Q. No.:		Probable Answer Keys		Write Your Answer Here
i.	Measure of cause and effect relationship	a)	Dispersion	i. -
ii.	Goodness of fit test	b)	1 - α	ii. -
iii.	Measure of the variation of the items	c)	Chi-square test	iii. -
iv.	Confidence efficient	d)	1 - β	iv. -
v.	Class frequency / Number of observations	e)	Class Distribution	v. -
vi.	Peakness of Curve	f)	Relative Frequency	vi. -
		g)	Kurtosis	
		h)	F-test	
		i)	Time Series Analysis	
		j)	Regression Analysis	

B) Multiple-Choice Questions (vii – ix) (Note: Encircle the correct answer)

03

vii. Probability sampling and random sampling are

- | | |
|---------------|----------------------|
| a) Anonymous | b) Different terms |
| c) Synonymous | d) None of the above |

viii. Coefficient of correlation will be always

- | | |
|-----------------|-----------------------|
| a) More than 0 | b) More than - 1 |
| c) Less than -1 | d) Between -1 and + 1 |

- ix.** The median of 10, 12, 14, 16, 18 will be
- | | |
|-------|-------|
| a) 12 | b) 16 |
| c) 15 | d) 14 |

C) Fill-in the Blanks (x – xii) (You may choose from the following answers):

03

Outliers	Type-I error	04	Central liners	20	Type-II error
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- x.** Probability of rejecting the null hypothesis when it is true is called_____.
- xi.** In a set of observations, unusual lower and higher values are called_____.
- xii.** Largest value is 60 and smallest value is 40 and number of classes desired is 5 then class interval is _____.

D) True or False (xiii – xv) (Note: Encircle the correct answer)

03

- xiii.** In arithmetic mean, sum of deviations of all recorded observations must always be zero.

(True) or (False)

- xiv.** Sum of values of data is divided by total number of values is used to calculate geometric mean.

(True) or (False)

- xv.** Summary and presentation of data in tabular form with several non-overlapping classes is referred as frequency distribution.

(True) or (False)

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BBA PROGRAMME

SEMESTER – II, UNIVERSITY EXAMINATION, SEPTEMBER - 2018

BM110 BUSINESS STATISTICS (OLD)

BM110.1 BUSINESS STATISTICS (NEW)

Date: 20/09/2018, Thursday

Time: 10.30 AM to 01.00 PM

Total Marks: 55

Note: - (i) All questions are compulsory.

(ii) This is a closed book examination.

(iii) There are two sections in the question paper. **Section – I** contains Q. No.: 1 of 15 marks and **Section – II** contains total five questions where in Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.

(iv) **Section – I** should be answered in the given question paper and **Section – II** should be answered in a separate answer book.

(v) Maximum time allotted for **Section – II** is 150 minutes.

(vi) Figures on the right indicate marks for each question.

SECTION – II

Q.2 The data below shows Age (in years) of 100 people: **10**

34, 67, 40, 72, 37, 33, 42, 62, 49, 32, 52, 40, 31, 19, 68, 55, 57, 54, 37, 32,
54, 38, 20, 50, 56, 48, 35, 52, 29, 56, 68, 65, 45, 44, 54, 39, 29, 56, 43, 42,
22, 30, 26, 20, 48, 29, 34, 27, 40, 28, 45, 21, 42, 38, 29, 26, 62, 35, 28, 24,
44, 46, 39, 29, 27, 40, 22, 38, 42, 39, 26, 48, 39, 25, 34, 56, 31, 60, 32, 24,
51, 69, 28, 27, 38, 56, 36, 25, 46, 50, 36, 58, 39, 57, 55, 42, 49, 38, 49, 36

i. Summarize the data in a frequency table. (with class intervals starting with 18 – 22, 23 – 27.....)

ii. **Calculate:**

- a. Cumulative Frequency
- b. Relative Frequency
- c. % Frequency
- d. Mean

OR

Q.2 Following are the marks out of 100 obtained by the students in a Statistics test: **10**

70	55	51	42	57	45	60	47	63	53	33	65
39	82	55	64	50	25	65	75	30	20	58	52
36	45	42	35	40	61	53	59	49	41	15	52
46	42	45	39	55	65	45	63	54	48	64	35
26	18										

- i. Make a frequency distribution taking a class interval of 10 marks.
- ii. Draw a histogram and a frequency polygon for the frequency distribution.
- iii. Find out the average marks of the students.

Q.3 **Solve the following problems (A and B):** **10**

- A An investor is fond of investing in equity shares. During a period of falling prices in the stock exchange, a stock is sold at Rs 120 per share on one day, Rs 105 on the next and Rs 90 on the third day. The investor has purchased 50 shares on the first day, 80 shares on the second day and 100 shares on the third' day. What average price per share did the investor pay? [05]

- B The following data give the savings bank accounts balances of nine sample households selected in a survey. The figures are in rupees. [05]

745	2000	1500	68000	461	549	3750	1800	4795
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- Find the mean and the median for these data.
- Do these data contain an outlier? If so, exclude this value and recalculate the mean and median. Which of these summary measures has a greater change when an outlier is dropped?
- Which of these two summary measures is more appropriate for this series?

Q.4 Solve the following problems (A and B):

10

- A Following are the heights and weights of 10 students of a BBA class: [05]

Height (in inches) X	62	72	68	58	65	70	66	63	60	72
Weight (in Kgs.) Y	50	65	63	50	54	60	61	55	54	65

Draw a scatter diagram and indicate whether the correlation is positive or negative?

- B The Final exam scores in a statistics class were normally distributed with a mean of 63 and a standard deviation of 5. [05]

- Find the probability that a randomly selected student scored more than 65 in the exam.
- Find the probability that a randomly selected student scored less than 85.

OR

- Q.4** Suppose you design a research project that concerns all of the students of your university. Decide on a sample size and discuss how you could use the four non-probability sampling methods. Discuss the potential problems and possible solutions to the problems. **10**

- Q.5** A brochure inviting subscriptions for a new diet program states that the participants are expected to lose over 22 pounds in five weeks. Suppose that, from the data of the five-week weight losses of 56 participants, the sample mean and sample standard deviation are found to be 23.5 and 10.2, respectively. Could the statement in the brochure be substantiated on the basis of these findings? Test at the $\alpha = 0.05$ level. **10**

Q.6 Case-let(s)

15

- A Sixty percent of new drivers have had driver education. During their first year, new drivers without driver education have probability 0.08 of having an accident, but new drivers with driver education have only a 0.05 probability of an accident. What is the probability a new driver has had driver education, given that the driver has had no accident the first year? 05

- B A pair of dice consisting of a six-sided die and a four-sided die is rolled and the sum is determined. Let the random variable X denote this sum. Find the sample space, the space of the random variable, and probability density function of X. 05

- C Figures released by the U.S. Department of Agriculture show that the average size of farms has increased since 1940. In 1940, the mean size of a farm was 174 acres; by 1997, the average size was 471 acres. Suppose an agribusiness researcher believes the average size of farms has now increased from the 1997 mean figure of 471 acres. To test this notion, a randomly sampled 23 farms across the US were selected and ascertained the size of each farm from county records. The data gathered is follow. Use a 5% level of significance to test the hypothesis. Assume that number of acres per farm is normally distributed in the population. 05

445	489	474	505	553	477	454	463	466
557	502	449	438	500	466	477	557	433
545	511	590	561	560				